

Proof. **1. Inner-product expansion.** By linearity in the first argument and conjugate symmetry,

$$\begin{aligned}\langle x + y, x + y \rangle &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2.\end{aligned}$$

In a real inner-product space $\operatorname{Re} \langle x, y \rangle = \langle x, y \rangle$; in the complex case the argument below is unchanged because $|\langle x, y \rangle| \geq \operatorname{Re} \langle x, y \rangle$.

2. Cauchy–Schwarz inequality.

$$|\langle x, y \rangle| \leq \|x\| \|y\|,$$

with equality iff x and y are linearly dependent.

3. Bounding the mixed term. Since $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$,

$$2 \operatorname{Re} \langle x, y \rangle \leq 2 |\langle x, y \rangle| \leq 2 \|x\| \|y\|.$$

4. Inequality for the squared norm. Insert the estimate from step3 into the expansion of step1:

$$\begin{aligned}\|x + y\|^2 &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2 \\ &\leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2.\end{aligned}$$

5. Taking square roots. Norms are non-negative, hence the square-root function is monotone on $[0, \infty)$. From the previous inequality we obtain

$$\|x + y\| \leq \|x\| + \|y\|.$$

6. Degenerate cases. If $x = 0$ (or $y = 0$) the inequality reduces to $\|y\| \leq \|y\|$ (or the symmetric statement), which is true. When both vectors are zero both sides equal 0.

7. Equality condition. Equality in step4 requires equality in both inequalities of step3:

[(a)]

1. $\operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|$ forces $\langle x, y \rangle$ to be a non-negative real number.
2. $|\langle x, y \rangle| = \|x\| \|y\|$ forces linear dependence, i.e. there exists $\alpha \in \mathbb{C}$ with $y = \alpha x$.

Writing $y = \alpha x$ we have $\langle x, y \rangle = \alpha \|x\|^2$. Condition (a) then gives $\alpha \|x\|^2 \in [0, \infty)$, so α is a non-negative real scalar. Setting $\lambda := \alpha \geq 0$ yields $y = \lambda x$.

Conversely, if $y = \lambda x$ with $\lambda \geq 0$, then

$$\|x + y\| = \|(1 + \lambda)x\| = (1 + \lambda)\|x\| = \|x\| + \|\lambda x\| = \|x\| + \|y\|,$$

so equality holds. The cases $x = 0$ or $y = 0$ correspond to $\lambda = 0$.

Hence equality occurs exactly when the two vectors are positively collinear, i.e. there exists $\lambda \geq 0$ such that $y = \lambda x$. $\square$